32×32 Approximate CAM with Tunable Hamming Distance

User Manual

IHP SG13CMOS5L · VDD = 1.2 V · VEVAL_IN = 0.5 V · nominal clock 100 MHz, period $T = 10$ ns

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1 What the chip does

The chip stores up to 32 words of 32 bits. When it receives a search word, it compares it with all 32 stored words at once and returns one bit per word telling whether it is similar or not: **match** if the number of differing bits does not exceed a threshold, and **mismatch** if it does. The threshold is not fixed: the user programs it by writing a word into a special row called the *reference*.

Capacity	32 words × 32 bits
Comparison	Hamming distance against a programmable threshold
Interface	serial: 2 data inputs and 1 output, plus 3 operation pulses
Supplies	VDD = 1.2 V, VSS = 0 V, VEVAL_IN = 0.5 V
Clock	100 MHz nominal, rising edge
Writing one word	34 cycles, 340 ns
Programming the reference	34 cycles, 340 ns
Search with result readout	69 cycles, 690 ns

Table 1: Main characteristics.

2 Pins

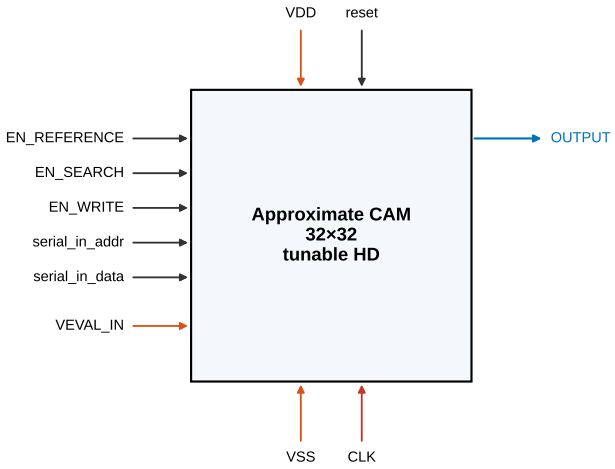


Figure 1: Chip symbol.

Pin	Type	Active	Function
`VDD`	Supply	—	1.2 V for the logic and the core.
`VSS`	Supply	—	Ground, 0 V.
`VEVAL_IN`	Analog input	—	0.5 V evaluation level. Sets how strongly the match lines discharge during a search.
`CLK`	Input	Rising edge	System clock, 100 MHz maximum.
`reset`	Input	High	Returns the chip to idle and clears the registers.
`EN_WRITE`	Input	High pulse	Writes one word into a row.
`EN_REFERENCE`	Input	High pulse	Programs the reference, that is, the threshold.
`EN_SEARCH`	Input	High pulse	Searches a word and returns the result on `OUTPUT`.
`serial_in_addr`	Input	Serial	5-bit row address, bit 4 first.
`serial_in_data`	Input	Serial	32-bit word, bit 31 first.
`OUTPUT`	Output	Serial	Search result, one bit per row: 0 = match, 1 = mismatch.

Table 2: Chip pins. Logic signals are CMOS, 0 to 1.2 V.

Data are always sent in their normal logic form. The chip internally adjusts the polarity the cells need, so the user does not have to invert anything.

3 Power-up

1. Apply `VDD` = 1.2 V and `VEVAL_IN` = 0.5 V, with `VSS` grounded.
2. Start `CLK` at 100 MHz or less, with a 50 % duty cycle.
3. With every `EN_*` at 0, hold `reset` at 1 for at least 2 clock cycles and then release it.
4. Program the reference with the desired threshold (Section 5).
5. Write the words to be used (Section 6).
6. Searches can now be performed (Section 7).

After a reset or at power-up, the content of the rows and of the reference is unknown: they must be written again.

4 Timing rules

- The chip captures every input on the **rising edge**. The safest choice is to change inputs on the **falling edge**.
- An operation starts on the first rising edge at which the chip sees its `EN_*` high while idle. That edge is called **E0**, and the following ones E1, E2, and so on.
- A one- or two-cycle `EN_*` pulse is enough. Each pulse launches a single operation; to repeat it, the pin must go low and high again.
- Bit number k of a serial input, counting from $k = 0$, is captured on edge E$(k + 1)$.
- Between operations: the next one can start **35 cycles** after a write or reference, and **70 cycles** after a search. A pulse that arrives earlier is ignored; it is not queued.
- If several `EN_*` arrive together, the priority is write, then reference, then search.

5 Setting the threshold

During a search, the reference is always compared against the fixed pattern `0xFFFF0000`. The number of bits in which the reference word R differs from that pattern is called m_{ref}, and it sets the threshold:

$$\text{match} \iff \text{differing bits} \leq m_{\text{ref}} - 4.$$

To build R, start from `0xFFFF0000` and invert any m_{ref} bits.

m_{ref}	Word R	Match when differing bits are	Status
19	`0xFFF8FFFF`	**15 or fewer**	**verified, recommended setting**
20	`0xFFF0FFFF`	16 or fewer	verified: 16 is a match and 18 a mismatch
16	`0xFFFFFFFF`	12 or fewer	estimated
24	`0xFF00FFFF`	20 or fewer	estimated

Table 3: Reference words. Verified values come from simulation at the typical corner; on the fabricated chip they should be confirmed with the procedure of Section 9.

Procedure:

1. Pulse `EN_REFERENCE`.
2. Send the 32 bits of R on `serial_in_data`, from bit 31 to 0, captured from E1 to E32. `serial_in_addr` is not used.
3. The reference is stored at E34 and is kept for all following searches, until it is reprogrammed or a reset occurs.

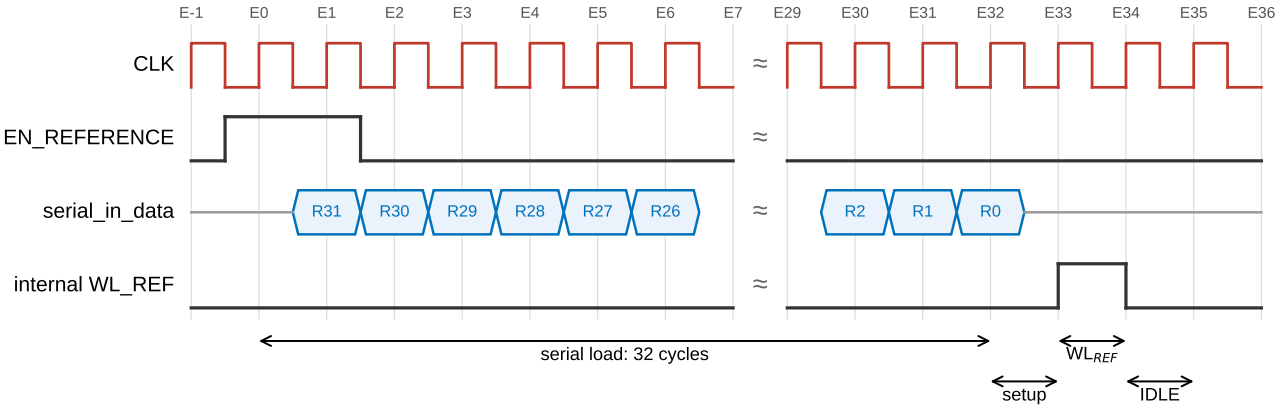


Figure 2: Reference programming protocol.

6 Writing a word

1. Pulse `EN_WRITE`.
2. Send the 32 bits of the word on `serial_in_data`, from bit 31 to 0, captured from E1 to E32.
3. At the same time, during the first five cycles, send the row address on `serial_in_addr`, from bit 4 to 0, captured from E1 to E5.
4. The word is stored at E34.

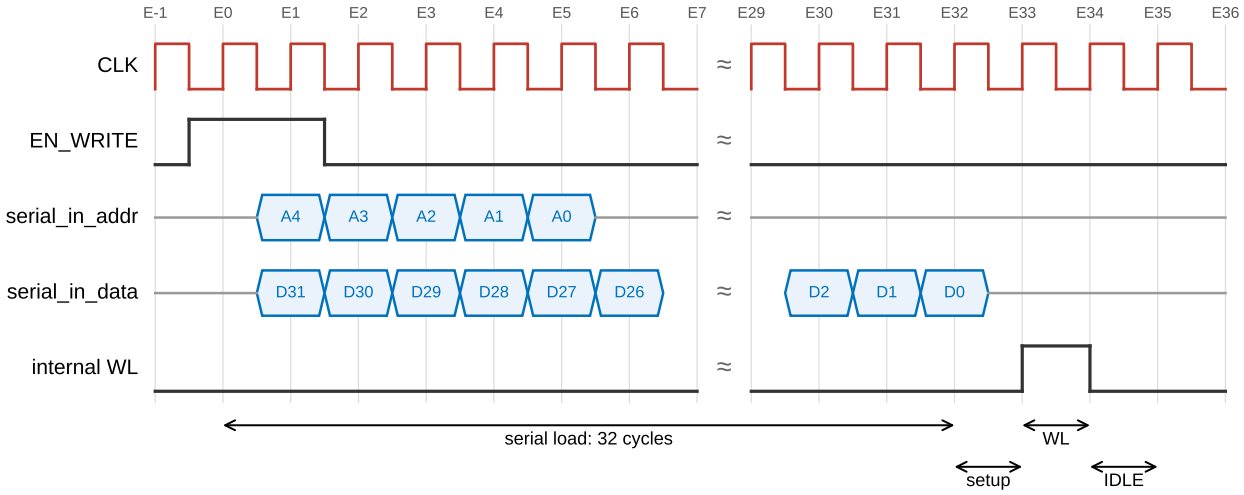


Figure 3: Write protocol. The *internal WL* signal is not a pin; it shows when the data is stored.

7 Searching and reading the result

1. Pulse `EN_SEARCH`.
2. Send the search word on `serial_in_data`, from bit 31 to 0, captured from E1 to E32.
3. Wait five cycles, from E32 to E37, while the chip compares the 32 rows and stores the result.
4. Read `OUTPUT` for 32 cycles, from E37 to E69. The bit of row i is valid between E$(68-i)$ and E$(69-i)$; it is best read on the falling edge of that cycle. Row 31 comes out first and row 0 last.
5. Interpret: **0 = match**, **1 = mismatch**. Bits of rows that were not written have no meaning.

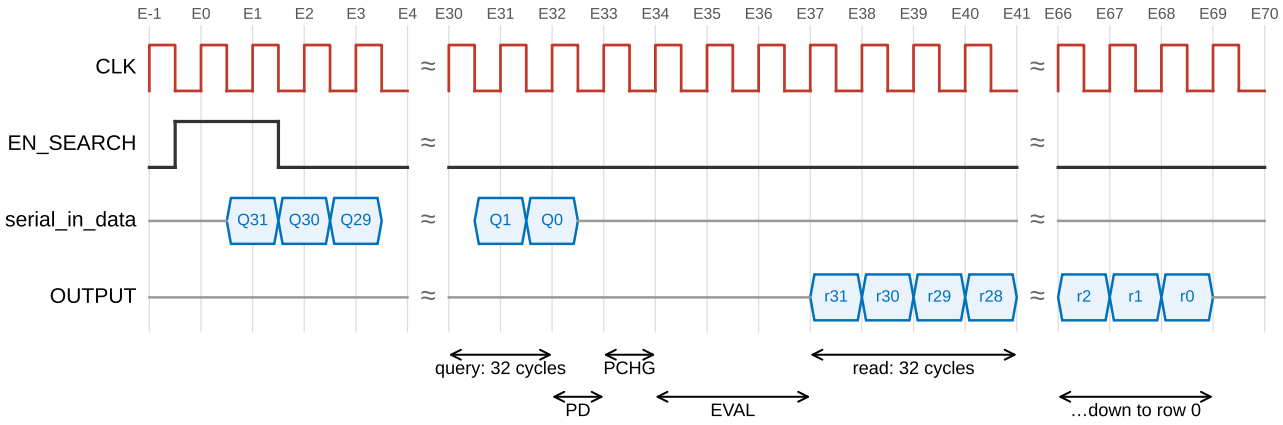


Figure 4: Search and readout protocol. r31 to r0 are the results of rows 31 to 0.

8 Verified example

This sequence was simulated completely at transistor level: six writes, the reference $R =$ `0xFFF8FFFF` with $m_{\text{ref}} = 19$, and a search for $Q =$ `0x0000FFFF`. It is also a good first test for the fabricated chip.

Row	Address	Written word	Differing bits with Q	Expected `OUTPUT`
0	`00000`	`0x0000FFFF`	0	0, match
5	`00101`	`0x0000E000`	13	0, match
13	`01101`	`0x00FF01FF`	15	0, match
20	`10100`	`0x00FF00FF`	16	1, mismatch
22	`10110`	`0x00030000`	18	1, mismatch
31	`11111`	`0xFFFF0000`	32	1, mismatch

Table 4: Test sequence and expected result.

9 Calibration on the fabricated chip

The threshold comes from comparing discharge times, so on silicon it may shift by one or two bits with respect to simulation, depending on process, temperature and the value of `VEVAL_IN`. To set it precisely:

1. Choose a word Q and write several rows with known distances to Q, for example from 12 to 20 differing bits, one distance per row.
2. Program the reference with $m_{\text{ref}} = 19$ and search for Q.
3. Find the row at which `OUTPUT` changes from 0 to 1. That boundary is the real threshold.
4. If it is not where it is needed, raise or lower m_{ref} one step at a time and repeat. Each unit of m_{ref} moves the threshold by about one bit.

`VEVAL_IN` also has an effect: a higher value speeds up every discharge. Keeping it at 0.5 V and tuning the threshold only through the reference is recommended; if `VEVAL_IN` is changed, calibrate again.

10 Recommendations

- Launch one operation at a time and respect the minimum time between operations.
- Program the reference after every power-up or reset, and whenever the threshold needs to change.
- Do not exceed 100 MHz: the internal precharge and evaluation phases last a fixed number of cycles.
- Read `OUTPUT` on the falling edge, because it is a combinational output and may show short glitches right after each rising edge.